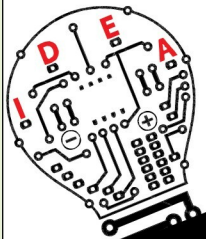


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Compatibility with other toolboxes

BioSig is not a single open source toolbox available for biomedical signal processing. Unlike in the case of proprietary software, the open source software are interdependent. The functionalities can be shared between various toolboxes and redundancy of work can be avoided.

For example, software like FieldTrip and SPM make use of BioSig for specific tasks. EEGLab, a widely-used toolbox in MATLAB for EEG and magnetoencephalography (MEG) signal processing, includes BioSig in it.

performing a single task. In order to study this connectivity and interaction, researchers make use of the auto-spectrum of a single channel and the cross spectrum of two different channels. Power of the cross spectrum when normalised with corresponding autospectra power is termed as coherence.

When we are dealing with the cross spectrum, coherence is one of the most commonly used measure. Coherence, along with other measures in studying brain interactions, could be easily estimated with the help of multi-variate autoregressive (MVAR) model. BioSig has a complete toolbox for MVAR modelling.

Brain-computer interfacing. A BCI system observes and analyses the activity of brain in order to recognise the intention of the user. Being a low-cost, portable, non-invasive method for studying brain functions, EEG signals are widely used in BCI systems. BioSig software contains many tools that facilitate BCI research.

The rtsBCI toolbox available with the package can be used for designing a system for online analysis. In order to characterise the spontaneous oscillatory EEG activity, spectral analysis is being used. The pre-processing of data helps in removing technical artefacts and the signals recorded from activities other than

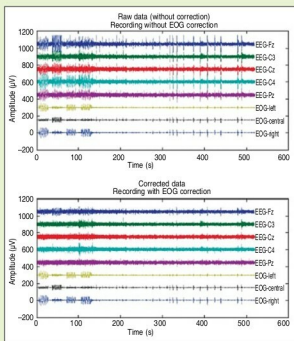


Fig. 2: Raw EEG data containing ocular artefacts can be rectified using regression analysis

that of the brain.

The use of special filters helps in concentrating on a particular part of the brain or a particular type of signal. Offline analysis proves to be helpful in many cases such as for estimation of a reliable classifier.

Nurturing biomedical research

The BioSig library consists of methods, algorithms and reference implementations for all kinds of biomedical signal-processing problems. If you are a fresher who expects a user-friendly environment, you might not find this tool very interesting. But, if you are a serious researcher or an engineer in search of a high-quality toolkit to aid your work, BioSig could be the ideal choice for you. ●

The author is an electronics enthusiast from Kerala